$P(\Delta < t_{\frac{\alpha}{2}} m)$

$F(138,999)$, $\Phi(x)$ - её ф-ия

Т14

$x \sim N(a, \sigma_x^2)$

$y \sim N(b, \sigma_y^2)$

$(\sigma_x^2, \sigma_y^2) = (2, 1)$

$H_0: a = b$

$H_1: a > b$

$x_n = \{-1,11; -6,10; 2,42\}$

$y_m = \{-2,29; -2,91\}$,

$\alpha = 0,05$

$D = \bar{x} - \bar{y}$

$\bar{x} \sim N(a, \frac{\sigma_x^2}{n})$.

$\bar{y} \sim N(b, \frac{\sigma_y^2}{m})$

$\Delta = \bar{x} - \bar{y} \sim N(a-b, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$

$$Z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}$$

$H_0: a - b = 0$

$Z \sim N(0; 1)$

$H_1: a > b$

$\bar{x} \approx -1,5967$

$\bar{y} = -2,6$

$\hat{\Delta} = \bar{x} - \bar{y} \approx 1,0033$

$r = \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}} = \sqrt{\frac{7}{6}} \approx 1,0801$

$\tilde{Z} = \frac{\hat{\Delta}}{r} \approx 0,9289$

$H_1: a > b$ – правост. альтерн.

$z_{1-\alpha} \approx 1,6449$

$G = \{Z > 1,6449\}$

p-value $\approx 0{,}1765 > \alpha$

$H_0$ - не отвергаем

• $\sigma = a - b$

$$w(\sigma) = P\left(\frac{\bar{x} - \bar{y}}{\sigma} > z_{1-\alpha} \mid H_1\right)$$

$$w(\sigma) = P\left(\frac{\bar{x} - \bar{y} - \sigma}{\sigma} > z_{1-\alpha} - \frac{\sigma}{\sigma} \mid H_1\right)$$

$$\left(\frac{\bar{x} - \bar{y} - \sigma}{\sigma}\right) \sim N(0,1)$$

$$w(\sigma) = \Phi\left(\frac{\sigma}{\sigma} - z_{1-\alpha}\right) = 1 - \Phi\left(z_{1-\alpha} - \frac{\sigma}{\sigma}\right)$$